AKILAN P L

B.E. CSE (AI/ML) Student

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github.com/akilanpl

PROFILE

AI/ML engineering student specialising in production RAG systems and on-premises LLM deployment, with end-to-end experience building enterprise AI platforms at KONE. Strong Python and backend foundation; SIH 2024 Finalist; open-source contributor to PyTorch. Targeting Software Engineering / ML Engineering internships.

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern (Enterprise Innovation Team), *KONE* 12/2025 – 01/2026
Chennai, Tamil Nadu, India

- Built an on-premises RAG platform using Python, LangChain, and a FAISS vector store, enabling secure conversational access to enterprise knowledge across 100+ internal documents — achieving ~30% lower query latency and ~40% improvement in retrieval relevance vs. the baseline keyword-search system.
- Automated PDF ingestion & extraction pipeline for 100+ enterprise documents using Python (PyMuPDF, pdfplumber), cutting manual processing time by an estimated 80% and eliminating copy-paste errors across reporting workflows.

Campus Ambassador, *Perplexity.ai* 09/2025 – 11/2025
Chennai, Tamil Nadu, India

- Ran hands-on AI tool workshops for 100+ students, driving measurable adoption of AI-assisted research workflows on campus.

TECHNICAL CONTRIBUTIONS

PyTorch (Meta) Open Source Contribution — Serialization Security Documentation, *https://github.com/pytorch/pytorch/pull/172743*

- Identified and documented a serialization security risk in torch.export (torch.load with weights_only=False); proposed and documented safer loading paths (weights_only=True, safetensors format).
- Contribution merged into the PyTorch core repo, improving security guidance for millions of developers worldwide.

PROJECTS

Enterprise RAG Platform, *https://github.com/akilanpl/enterprise-rag-platform* 11/2025 – 01/2026

- Developed an on-prem RAG system for enterprise knowledge access, achieving ~30% lower query latency, ~40% improvement in retrieval relevance, and supporting ingestion/search across 100+ internal documents with zero external data exposure.

Offline Mapping & Navigation System (SIH 2024 FINALIST) 08/2024 – 04/2025

- Built an offline geospatial navigation pipeline using Python to detect highway vs service-road movement through tile maps, heading vectors, and coordinate clustering algorithms.

Climate Detection Using AI 08/2024 – 12/2024

- Built a CNN-based climate classification model with custom preprocessing and evaluation pipelines; published research.

SKILLS

Core Languages
Python (primary), SQL, JavaScript, Java, C

AI/ML
RAG, LangChain, FAISS, on-prem LLM deployment, document ingestion & retrieval, CNN, model evaluation, secure model loading (safetensors)

Cloud
AWS (EC2, S3, IAM, networking), Azure (data workflows, ML), Google AI Studio

Backend & Systems
Flask, REST APIs, Python scripting, API development, automation workflows, Git / GitHub

Data & ML
Data pipelines, feature engineering, geospatial data processing (OSMnx, tile maps), model experimentation.

EDUCATION

B.E. CSE (AI & ML), *Rajalakshmi Institute of Technology* 09/2023 – Present
Chennai, Tamil Nadu, India

- Finalist, Smart India Hackathon 2024 — Offline Mapping & Navigation System
- Co-author of 2 academic research papers in AI/ML - Published at IJITRA

CERTIFICATES

AI Agents Intensive Program - Google
5-day program on AI agents, LLM workflows, and hands-on AI Studio projects.

AWS Cloud Foundations – AWS Academy
Practical training in EC2, S3, networking, IAM, and core cloud concepts.

Azure for Data Science - Microsoft
Built data workflows across ingestion, preparation, analytics, and ML on Azure.